

## **IAM Project - Documentation for creating my Active Directory environment**

**Active Directory** is a centralized identity management system used to manage users, computers, groups, and access permission within an organization's network

- AD runs on Windows Server 2022
- Active Directory Domain Controller is the central server that manages all security, user access, and computer configurations within a Windows network

### **Overview Takeaways from this project -**

#### **On Hands understanding of an AD, Entra ID, and Intune environment**

- Installed & Configured an Active Directory Domain Controller
- Installed & Configured an Entra ID & Intune
- Created Users, Organizational Units (OUs), Security Groups, Computers, Group Policy Object (GPOs), RBAC,
- Network Protocol Configuration - Configuring an Internal Network
- Domain-Joined Workstation,
- IP address, DNS server

### **Setup**

- For this section of the project I will use a Virtual Box VM to run Windows Server, giving me access to Active Directory
- In a real world environment the data from my Entra ID environment can be synced
- For learning purposes I will create an Active Directory environment manually

### **Installation Process**

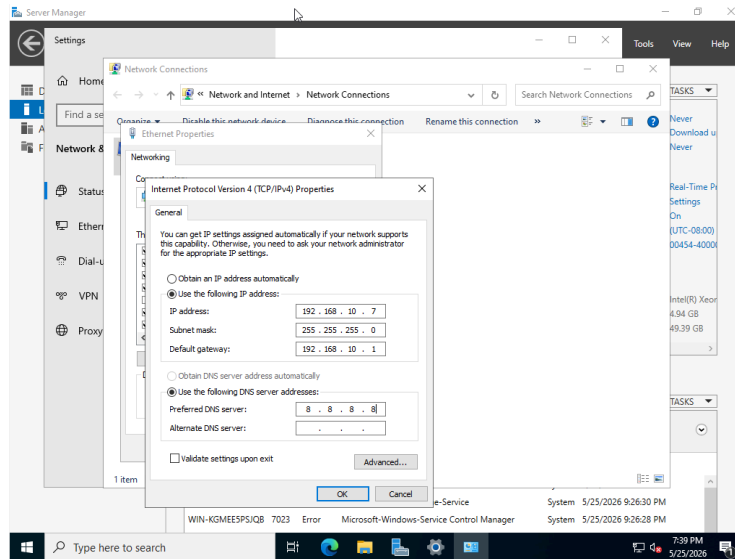
Since this lab is based on Active Directory and IAM environments, I will not document the installation process in detail. Here is an overview of the steps:

- Installed Virtual Box
- Configured Windows 10 Machine by installing Windows 10 ISO file and import within Virtual Box
- Configured Windows Server Machine by installing Windows Server ISO and import within Virtual Box
- Allocated proper RAM space, hardware specifications, etc. via VirtualBox

## Configuring Active Directory

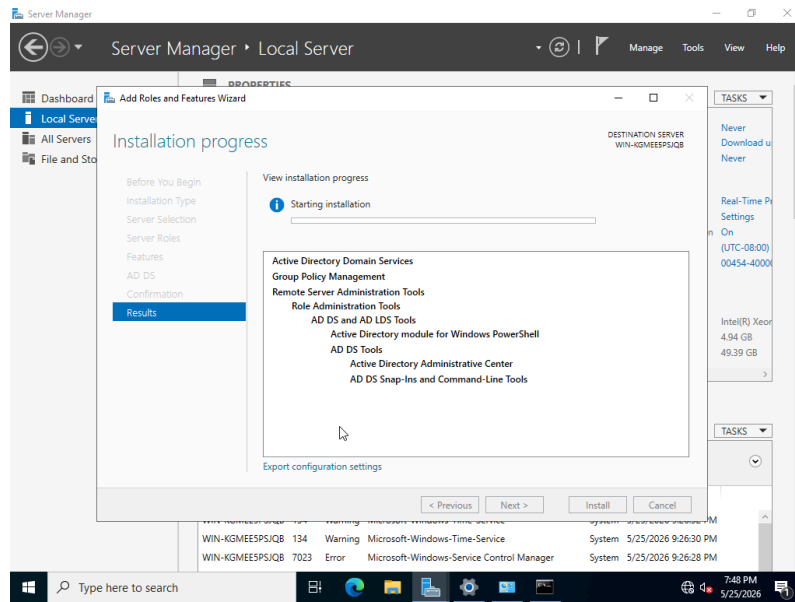
Created a Static IP

- Right Click Network Settings (Bottom Right Corner)
- Select Open Network & Internet Settings > Change adapter options> Right click “Ethernet” > Properties > Double click Internet Protocol Version 4
- This IP may be changed for future projects

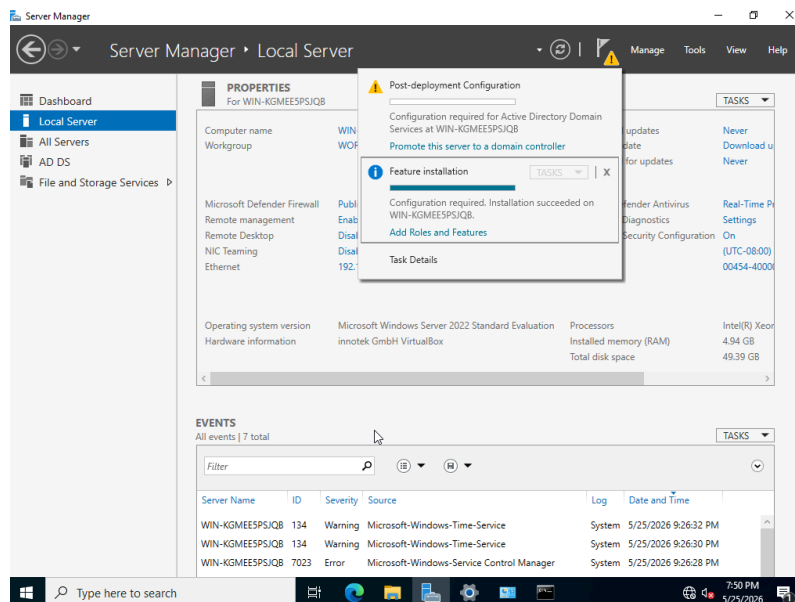


## Opening Active Directory

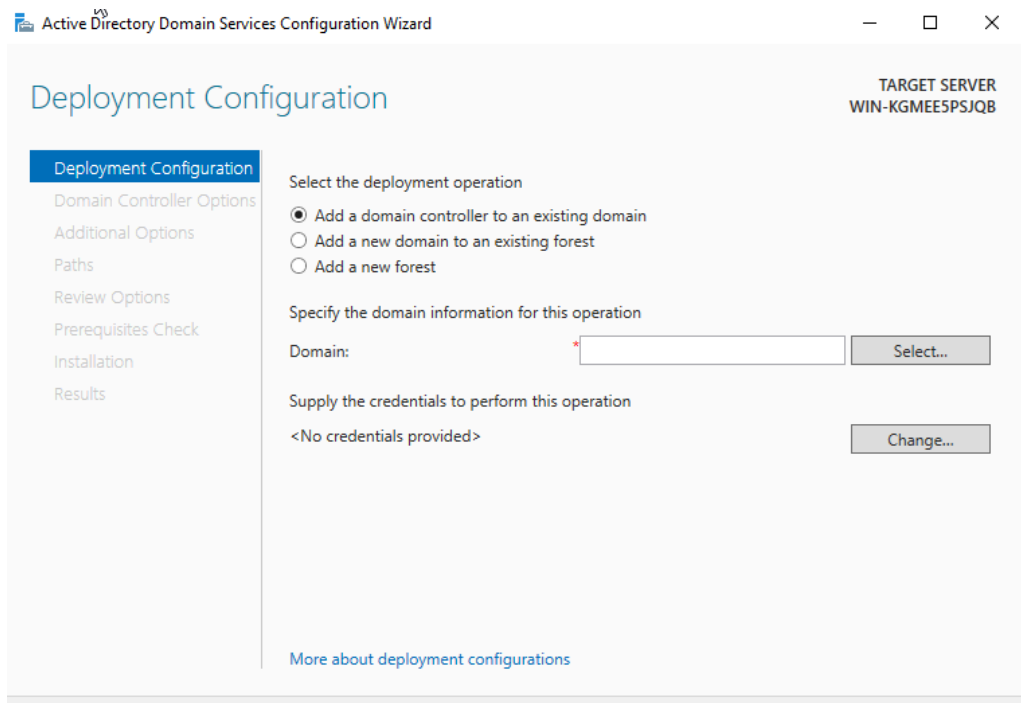
- Go into Server Manager
- Top right click Add Roles and Features > Next
- Installation Type: Make sure “Role-Based or feature-based Installation” is selected
- Server Selection: Next we will select the desired server (Windows 2022 VM)
- Server Roles: Active Directory Domain Services > Select Add Features
- Click next until you have the option to install



After installation we want to click the Flag (Top right) of the server manager to add the Domain Controller. Click “Promote this server to a domain controller” from there we will add the DC.

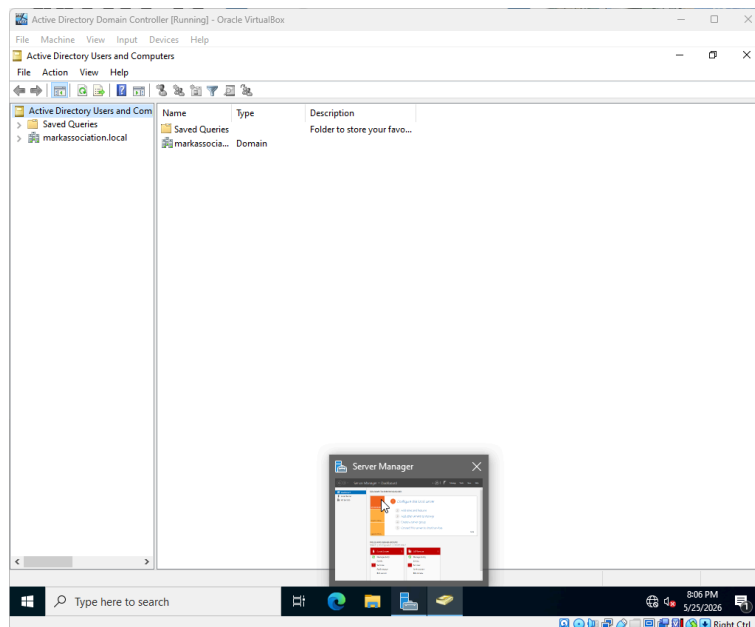


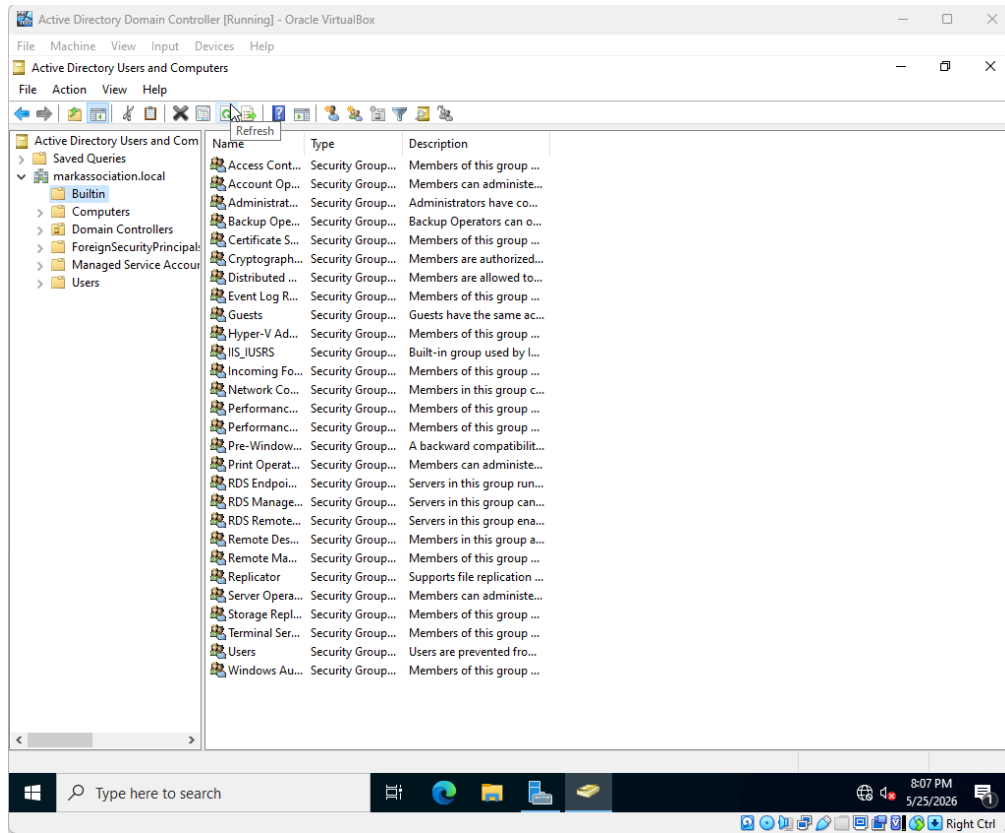
Since I am manually building my AD lab from scratch, I don't have an existing domain and I will add a new forest, then go through the configuration steps. Steps include creating a path to store the database, creating required passwords, NETBIOS name, and other information. Hackers like to attack Domain controllers because of the file in that path folder with user information and sensitive details.



After installation the VM restarted and successfully configured the Active Directory Domain Controller. Now I will start to create Users, OU's, Security Groups and establish my AD environment.

Once the Windows Server 2022 VM has related, open Server Manager > Tools (top right) > Active Directory Users and Computers. This is where I can create everything for my environment





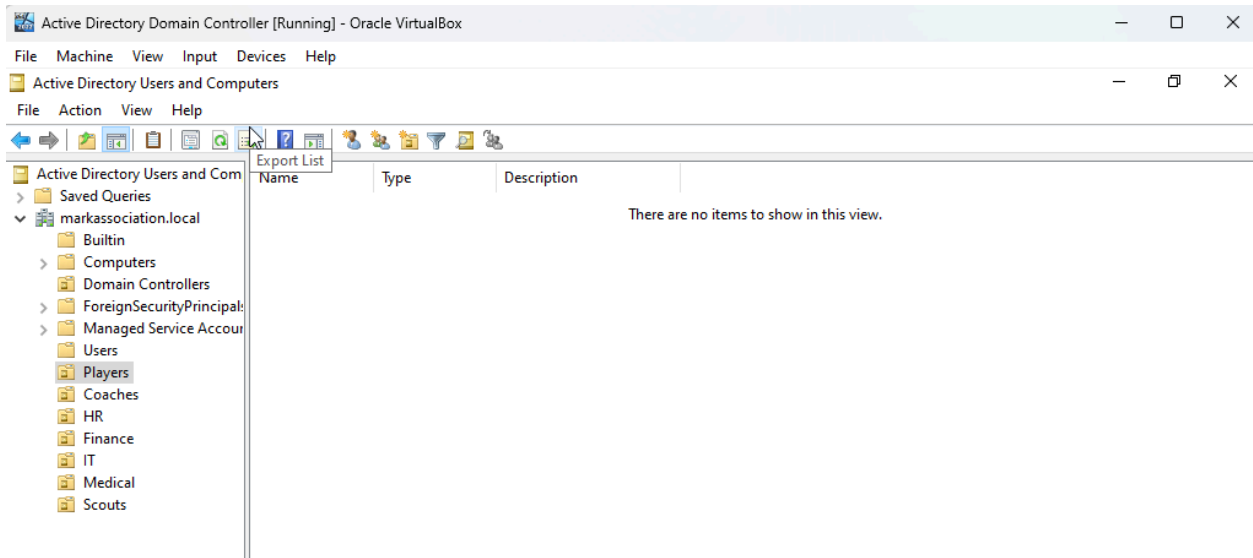
In the image above, you can see pre-built groups created in Active Directory, for example “Administrators”.

## Active Directory Environment

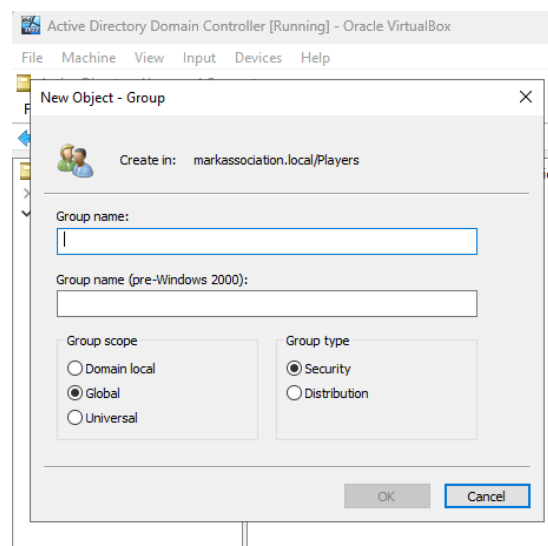
I will start by implementing my Organizational Units, Users, and Security Groups

- Right click domain > New > Organizational Unit
- Within that Unit I can create the new groups & users

## Organizational Units



On the left hand side I have initialized my **Organizational Units: Players, Coaches, HR, Finance, IT, Medical, Scouts**

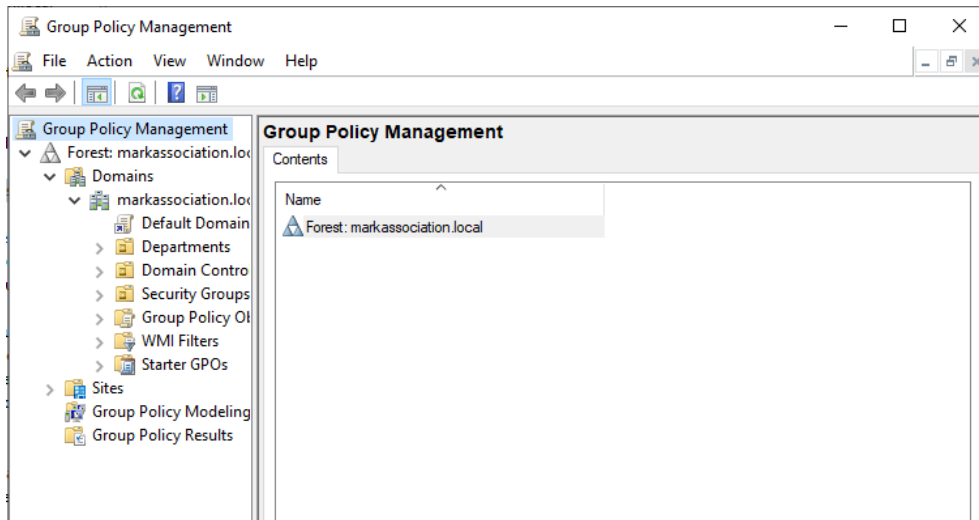


**Group Scope** - Where the group can be used

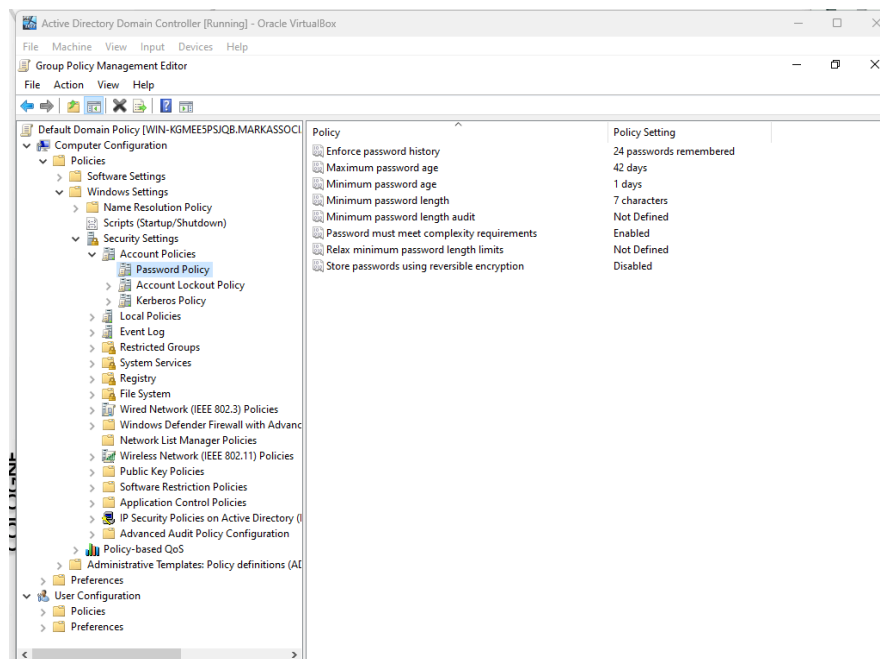
- Global - Most commonly selected, for Global access
- Domain Local - Usually used for assigning permissions to resources, for example my Playbook-view group.

You can also assign groups within a group, If Players and Coaches (Global Group Scope) need access to Playbook-View (Local Group Scope), they can be added

If you want to move OUs or groups make sure to right click folder > Properties > Object > Protect object from accidental deletion. Then u can move and delete the object, you have to do this everytime as a safety measuer



In order to change Group Policy information like password requirements I need to search in the Windows Bar > Group Policy Management > Open Forest domain > Domains > markassociation.local > Default Domain Policy > Right click, edit



For this lab I want the password to be simple so I have to change the **GPO**, after clicked edit on the default domain policy I need to go to Policies > Windows Settings > Security Settings > Account Policies > Password Policies, here I can disable the complex password requirement

| Policy                                      | Policy Setting          |
|---------------------------------------------|-------------------------|
| Enforce password history                    | 24 passwords remembered |
| Maximum password age                        | 42 days                 |
| Minimum password age                        | 1 days                  |
| Minimum password length                     | 4 characters            |
| Minimum password length audit               | Not Defined             |
| Password must meet complexity requirements  | Disabled                |
| Relax minimum password length limits        | Not Defined             |
| Store passwords using reversible encryption | Disabled                |

```

Administrator: Windows PowerShell
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> gpupdate /force
Updating policy...

Computer Policy update has completed successfully.

```

After changing the complexity I will open PowerShell and run a gpupdate /force. In the screen shot above you can see the successful update.

New Object - User

Create in: markassociation.local/Departments/Coaches

First name:

Initials:

Last name:

Full name:

User logon name:

@markassociation.local

User logon name (pre-Windows 2000):

MARKASSOCIATION\

< Back

Next >

Cancel

New Object - User

Create in: markassociation.local/Departments/Medical

Password:

....

Confirm password:

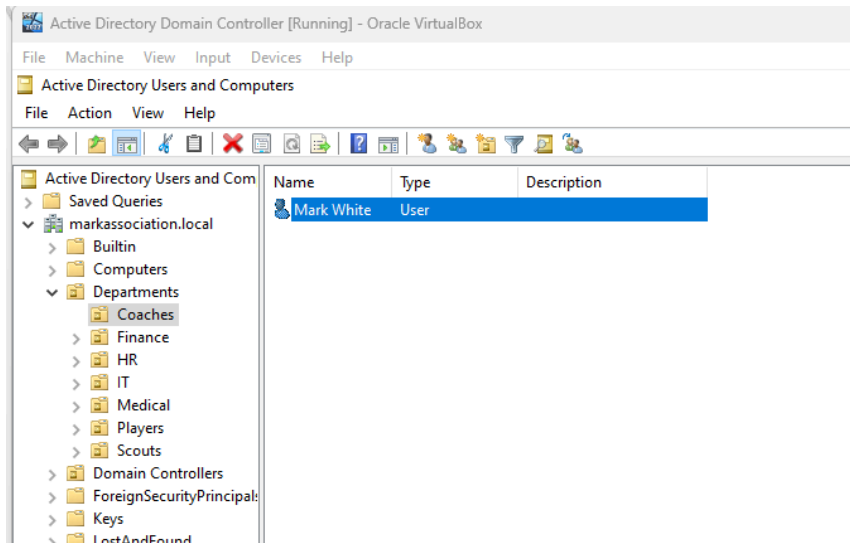
....

☒ User must change password at next logon
 ☐ User cannot change password
 ☐ Password never expires
 ☐ Account is disabled

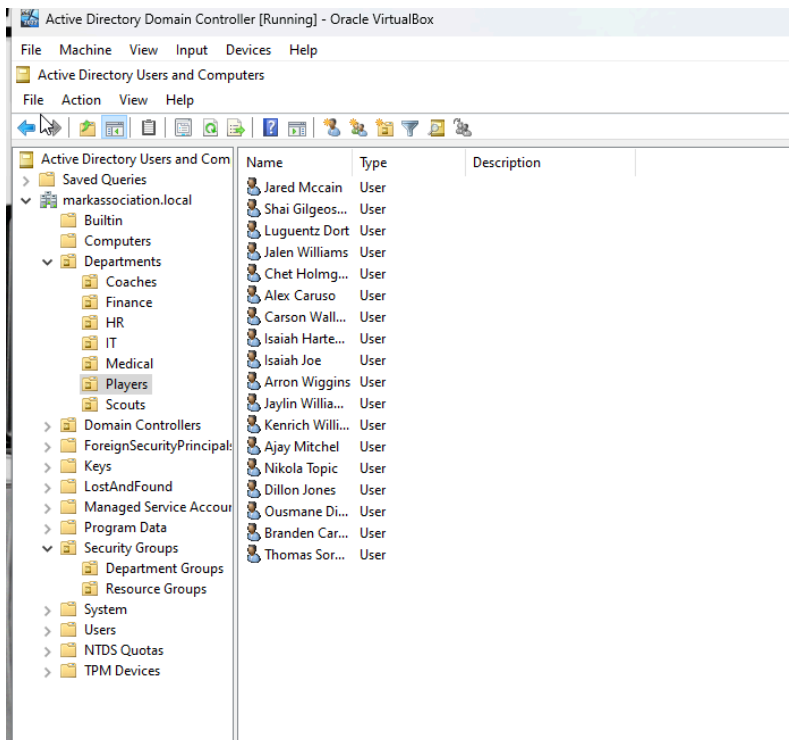
< Back

Next >

Cancel



Power shell can use a csv file to bulk add users, I will not be doing that for this process but in a real world scenario I would.



In the image above I have the OKC Roster added into the players OU, In the other OUs I just have one user created for each folder (Coaches, Finance, HR, Medical, Scouts)

Mark White Properties

Published Certificates Member Of Password Replication Dial-in Object  
Security Environment Sessions Remote control  
Remote Desktop Services Profile COM+ Attribute Editor  
General Address Account Profile Telephones Organization

 Mark White

First name:  Initials:

Last name:

Display name:

Description:

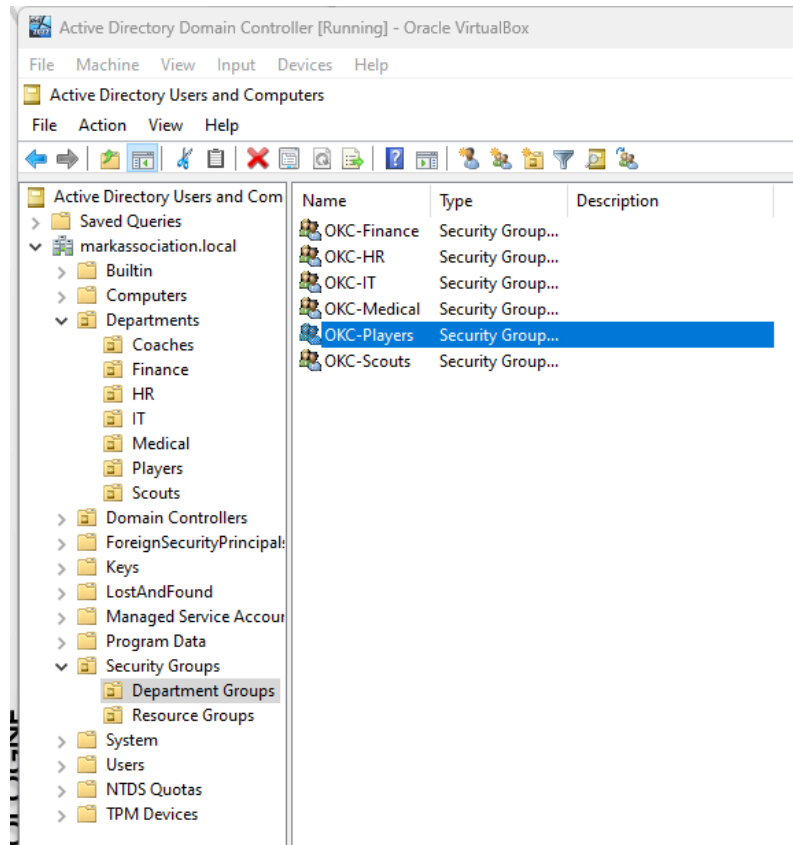
Office:

Telephone number:  Other...

E-mail:

Web page:  Other...

Now I have created a user, in the coaches Department Group. When creating a user I have the option to enter name, and login credentials. I will then be able to see the user's properties



Here is a look at my security groups, separated into Security Group > Department / Resource Group OUs. In order to automate this process a little bit, I will use PowerShell to implement my Resource groups a bit faster

```
Administrator: Windows PowerShell
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> hostname
WIN-KGME5PSJQB
PS C:\Users\Administrator> Get-ADDomain

AllowedDNSSuffixes      : {}
ChildDomains            : {}
ComputersContainer      : CN=Computers,DC=markassociation,DC=local
DeletedObjectsContainer : CN=Deleted Objects,DC=markassociation,DC=local
DistinguishedName       : DC=markassociation,DC=local
DNSRoot                 : markassociation.local
DomainControllersContainer : OU=Domain Controllers,DC=markassociation,DC=local
DomainMode              : Windows2016Domain
DomainSID               : S-1-5-21-1040356999-1770504990-220910587
ForeignSecurityPrincipalsContainer : CN=ForeignSecurityPrincipals,DC=markassociation,DC=local
Forest                  : markassociation.local
InfrastructureMaster     : WIN-KGME5PSJQB.markassociation.local
LastLogonReplicationInterval :
LinkedGroupPolicyObjects : {CN={31B2F340-016D-11D2-945F-00C04FB984F9},CN=Policies,CN=System,DC=markassociation,DC=local}
LostAndFoundContainer    : CN=LostAndFound,DC=markassociation,DC=local
ManagedBy               :
Name                     : markassociation
NetBIOSName              : MARKASSOCIATION
ObjectClass               : domainDNS
```

The two commands above verify I am on the correct Domain Controller to implement objects into my Active Directory environment

In order to create a group in Power Shell I will use:

```
New-ADGroup -Name "OKC-Coaches" -GroupScope Global -GroupCategory Security -Path
"OU=Resource Groups,OU=Security Groups,DC=markassociation,DC=local"
```

If I wanted to create multiple at once for my security groups I will use:

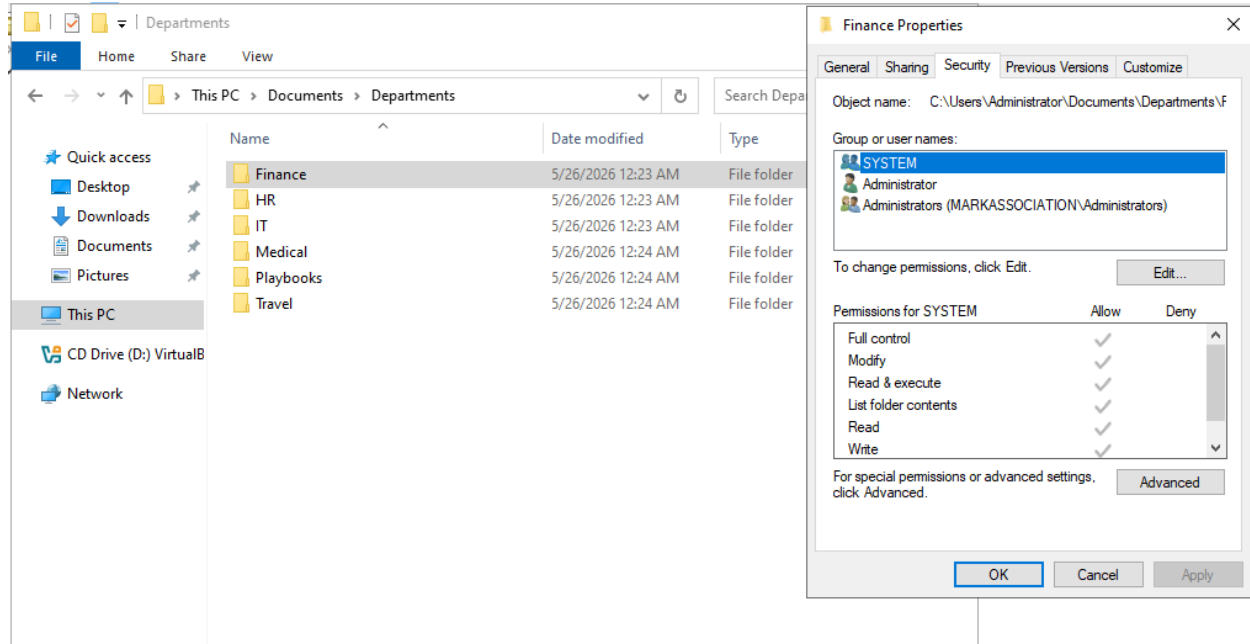
```
$groups = @(
"Playbook-View",
"Playbook-Editors",
"Finance-Reports",
"HR-Management",
"Media-Assets",
"Medical-Assets",
"Travel-Coordination"
)
```

```
foreach ($group in $groups) {
    New-ADGroup `
    -Name $group `
    -GroupScope Global `
```

```

-GroupCategory Security `
-Path "OU=Resource Groups,OU=Security Groups,DC=markassociation,DC=local"
}

```

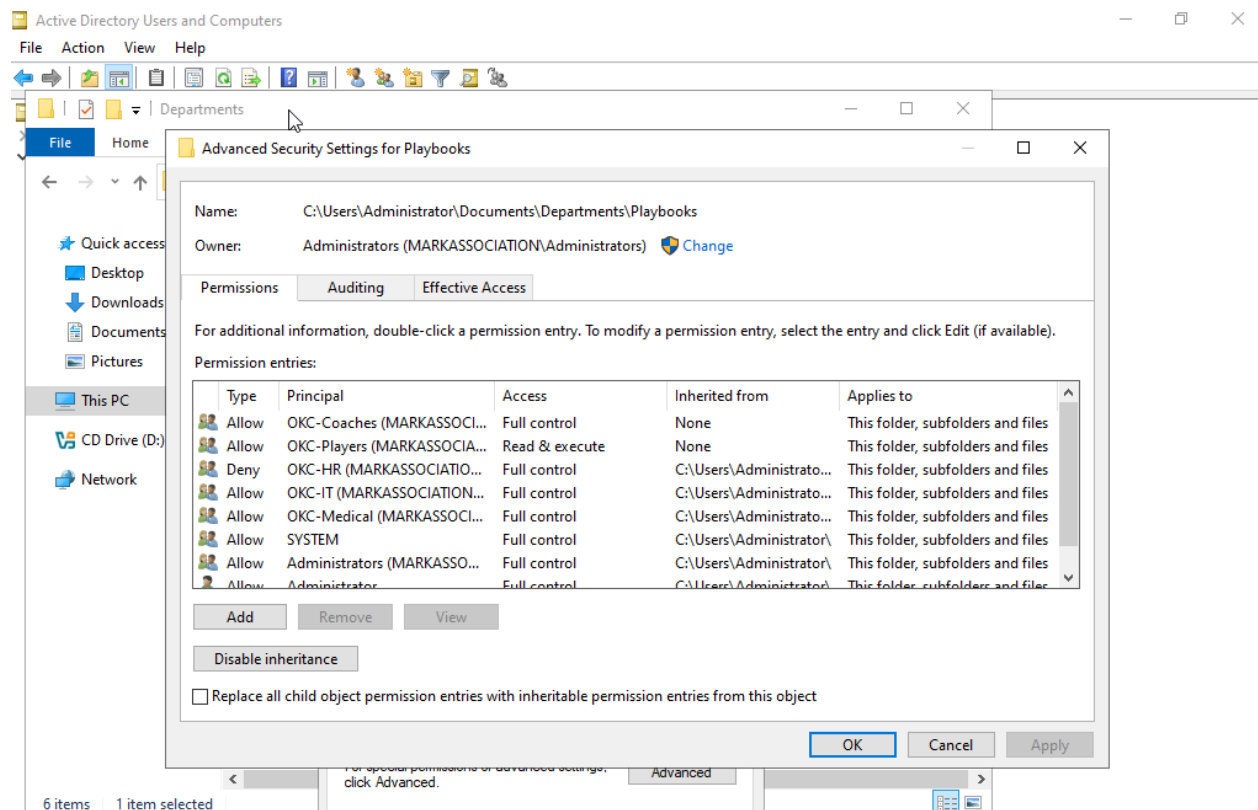


After creating the Security Groups, I will create folder and establish permissions where only specific groups can open specific folders. I can add restrictions to the folder by right clicking it > Properties > Security > Edit > Here I can add and remove groups, I will keep administrators and add the specific group names for each folder. From here I can also verify If I want that group to have Read Access, Full control, modify, etc

If inheritance is enabled, permissions from higher folders may still apply.

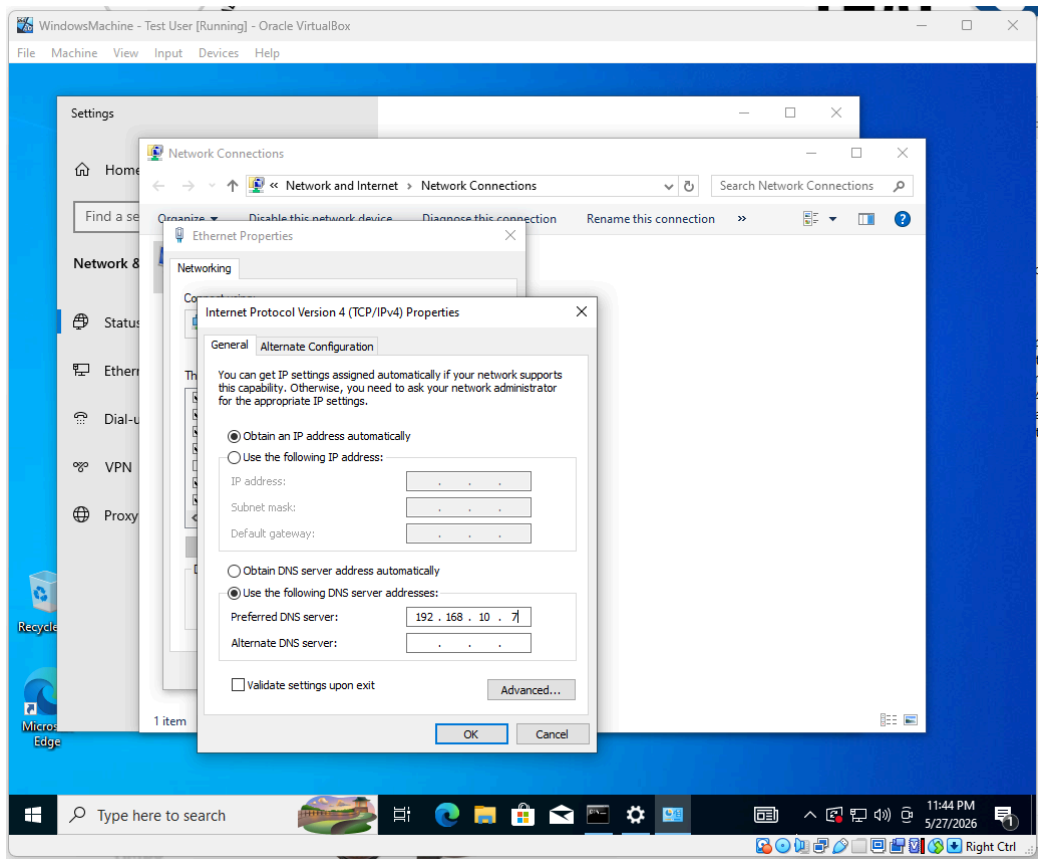
For tighter lab control, many admins:

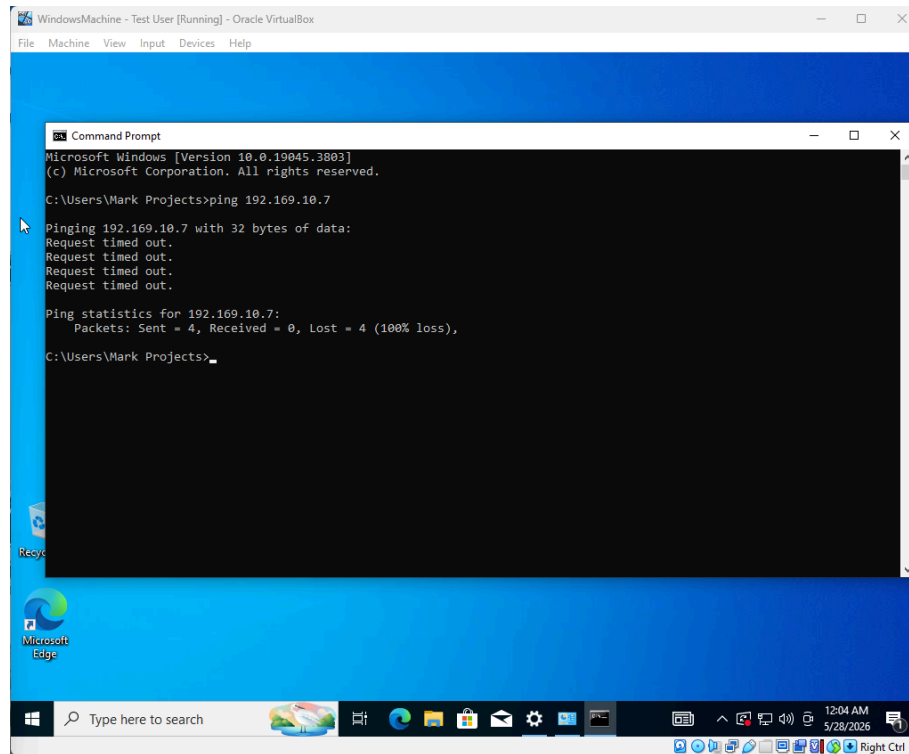
1. Disable inheritance
2. Convert inherited permissions
3. Remove unnecessary groups



In order to disable inheritance right click the folder > properties > Security > Advanced > Disable Inheritance

Next, I am going to add my Windows 10 VM to the Domain Controller under the computer OU. This VM will act as a user, inside my VM I will use PowerShell to find the specific Windows VM name, and add it to the domain controller.





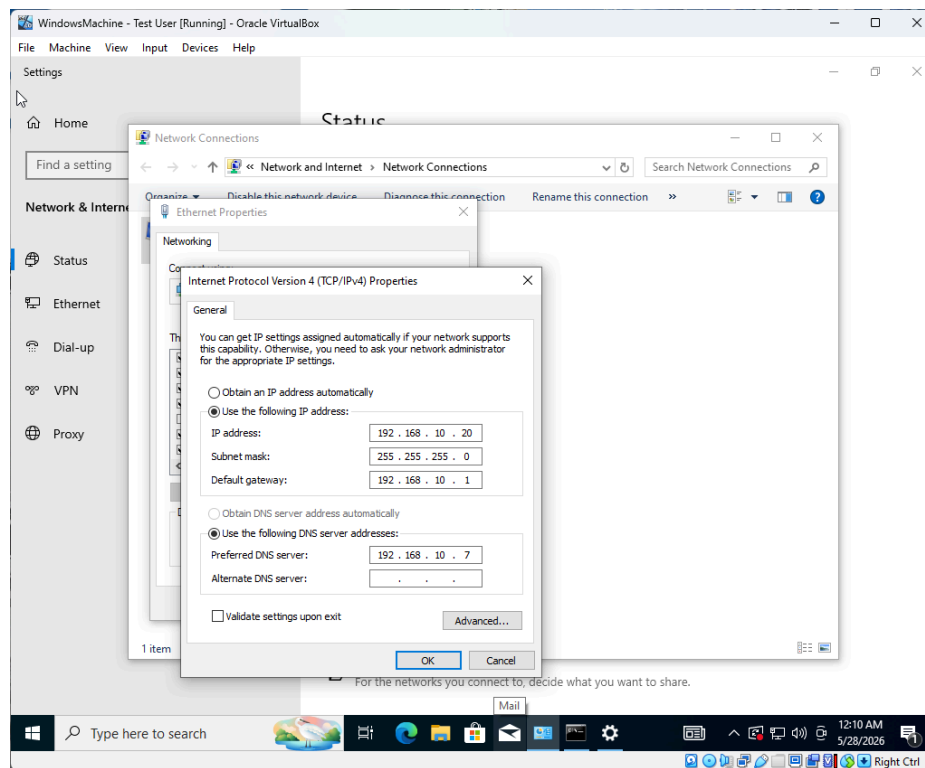
In order to verify the connection I will open CMD and ping the DC ip. As you can see the first time I tried (image above) the ping failed. After running ip config on my client VM I notice that the VM is on a different network

Active Directory DC ip address: 192.168.10.1

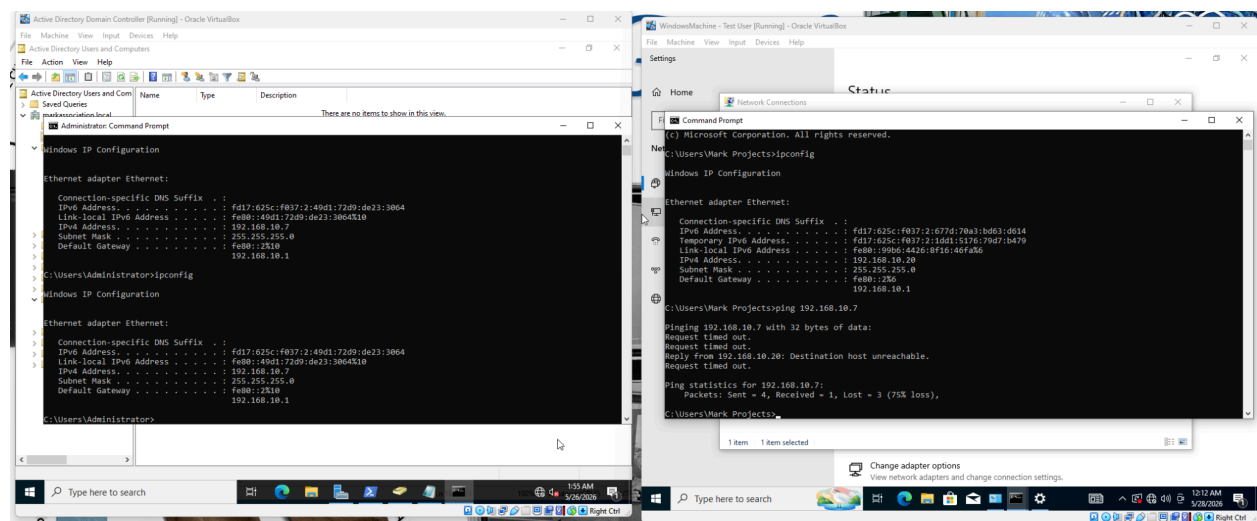
Windows 10 Client ip address: 10.0.2.15 -> converting this ip too 192.168.10.20

Gateway 192.168.10.1

DNS 192.168.10.7



The image above shows the proper IP static configuration for the client. Verified through ipconfig



I did receive another error, what I need to do is change the network settings in the Virtual box to make sure there the network type is set to a NAT Network

Here is a quick refresher of the different Network Protocols:

NAT - VM can access the internet through the host machine, but does not communicate with other devices

NAT Network - Multiple VMs can talk to each other and access the network

Bridged Adapter - VM appears as a real device on your physical network

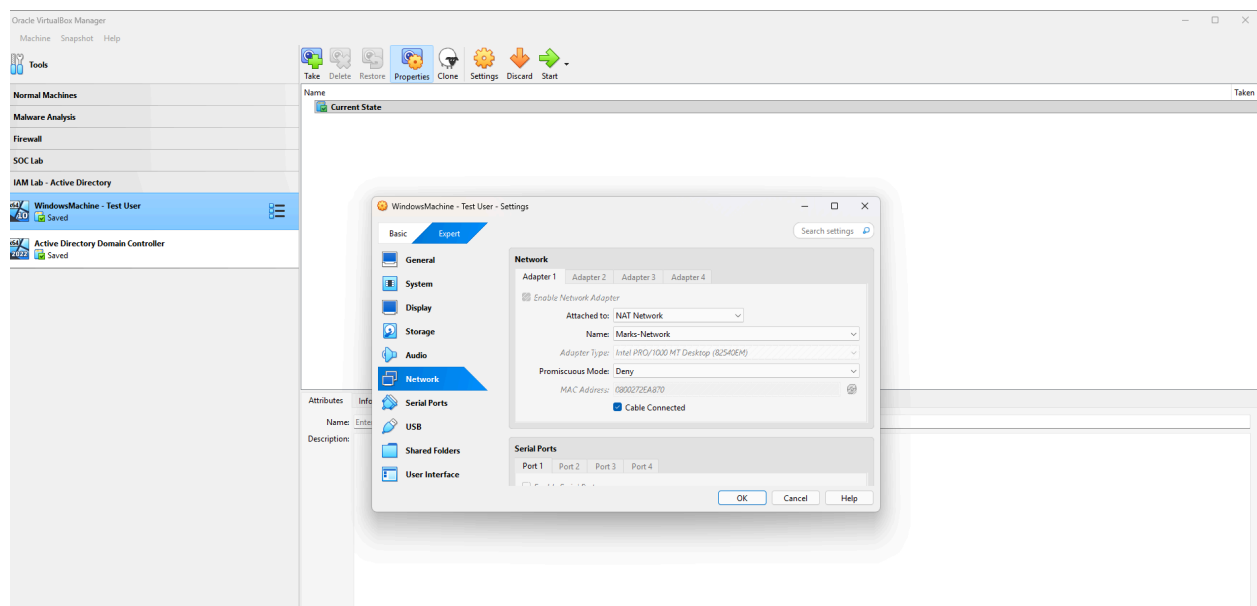
Host-only adapter - VMs can talk to each other and your host PC, but not the internet

Internal Network - VMs can only talk to other VMs on the same internal network

Not attached - Network cable unplugged, no connectivity

Internal Network - Works well for isolated labs

Host-Only Adapter -



In the image above I am configuring the Network protocols for my VMs. For this lab I will use a Internal Network protocol

```
Command Prompt
IPv4 Address. . . . . : 192.168.10.20
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::2%6
                          192.168.10.1

C:\Users\Mark Projects>ping 192.168.10.7

Pinging 192.168.10.7 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 192.168.10.20: Destination host unreachable.
Request timed out.

Ping statistics for 192.168.10.7:
    Packets: Sent = 4, Received = 1, Lost = 3 (75% loss),

C:\Users\Mark Projects>ping 192.168.10.7

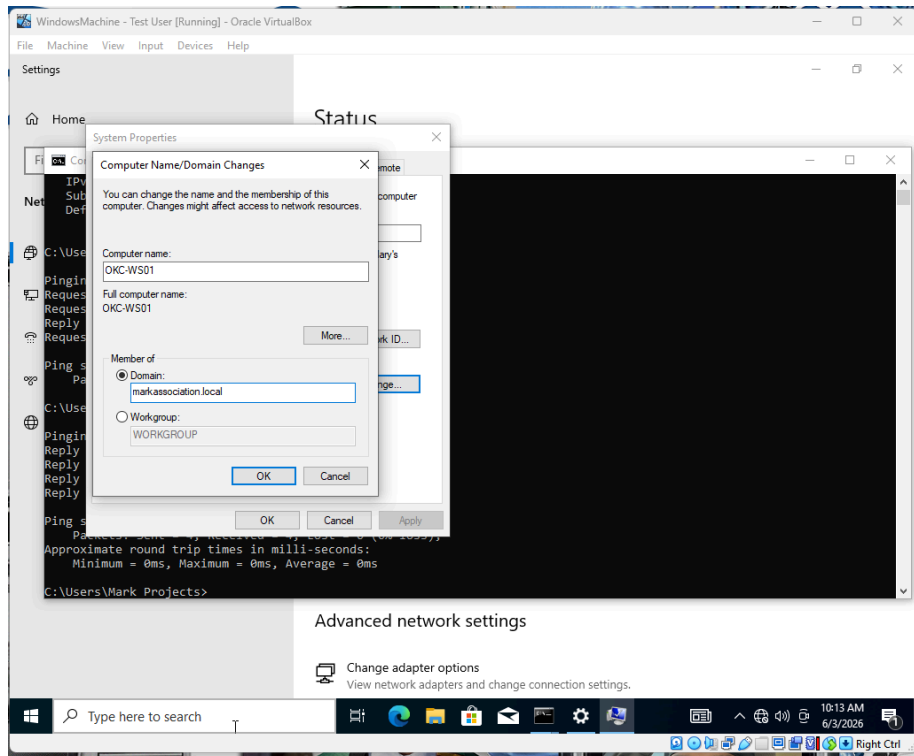
Pinging 192.168.10.7 with 32 bytes of data:
Reply from 192.168.10.7: bytes=32 time<1ms TTL=128
Reply from 192.168.10.7: bytes=32 time<1ms TTL=128
Reply from 192.168.10.7: bytes=32 time<1ms TTL=128
Reply from 192.168.10.7: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

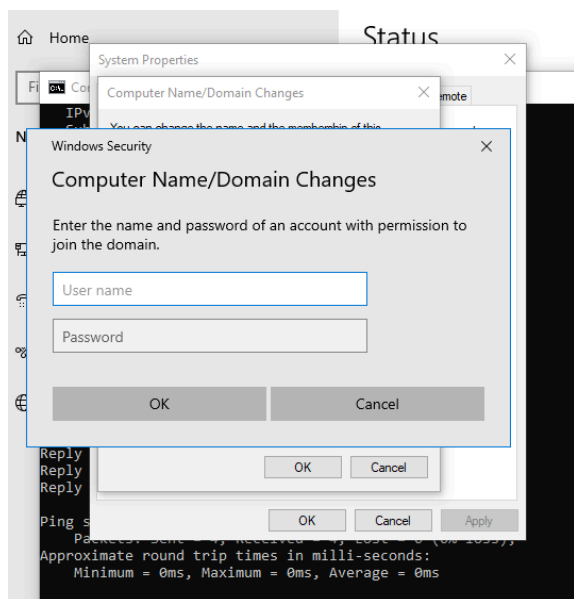
C:\Users\Mark Projects>
```

After configuration, I open my VM and ping the DC controller which is now receiving the packets

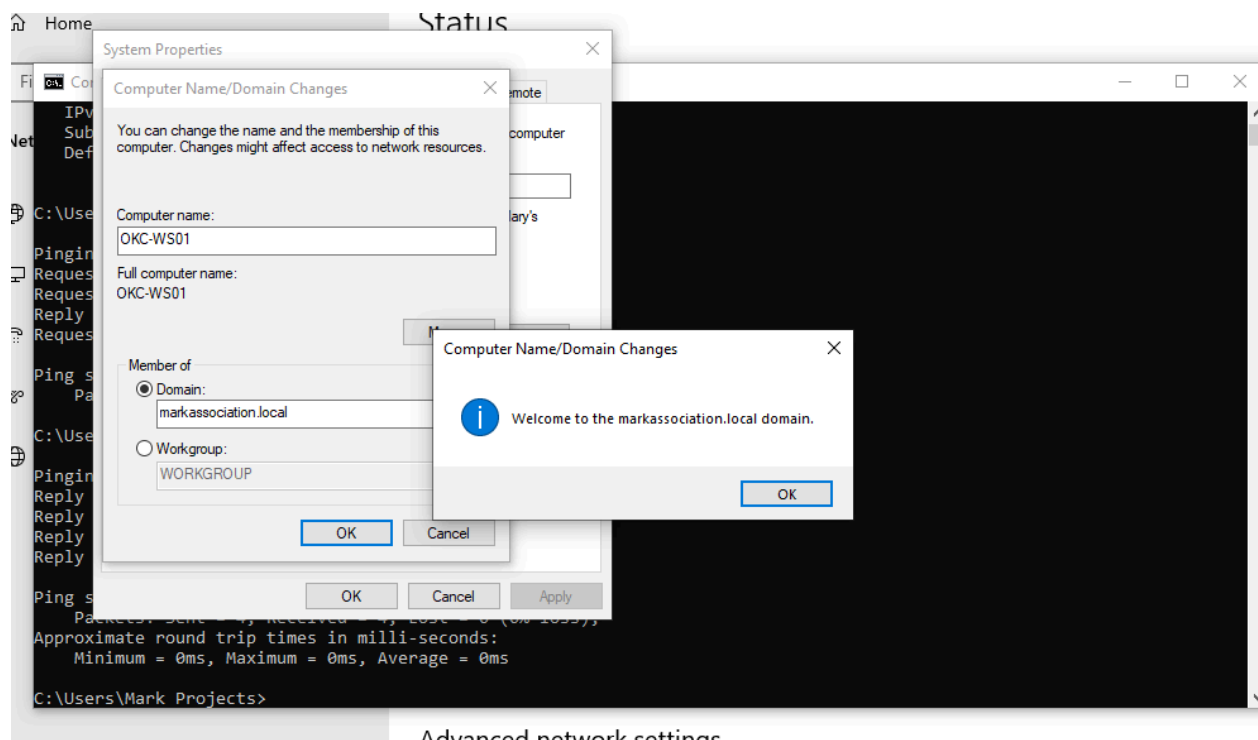
Now that we have configured the VM to be on the same network as the DC we're going to change the VM name & domain



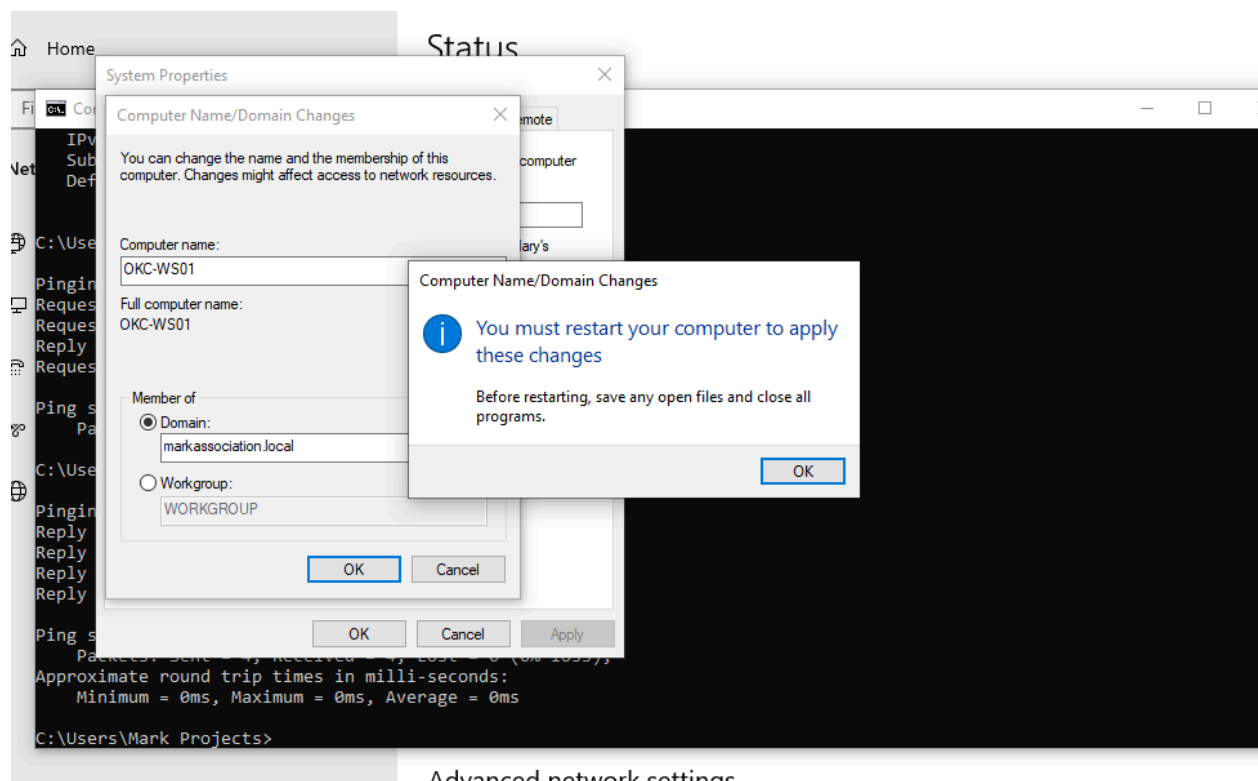
Search in Windows Bar sysdm.cpl → Computer Name → Change Domain → markassociation.local  
 Changing work station name to: OKC-WS01  
 Domain: markassociation.local



After clicking ok I will add:  
 User: markassociation\Administrator  
 Pass: <your DC Administrator password>

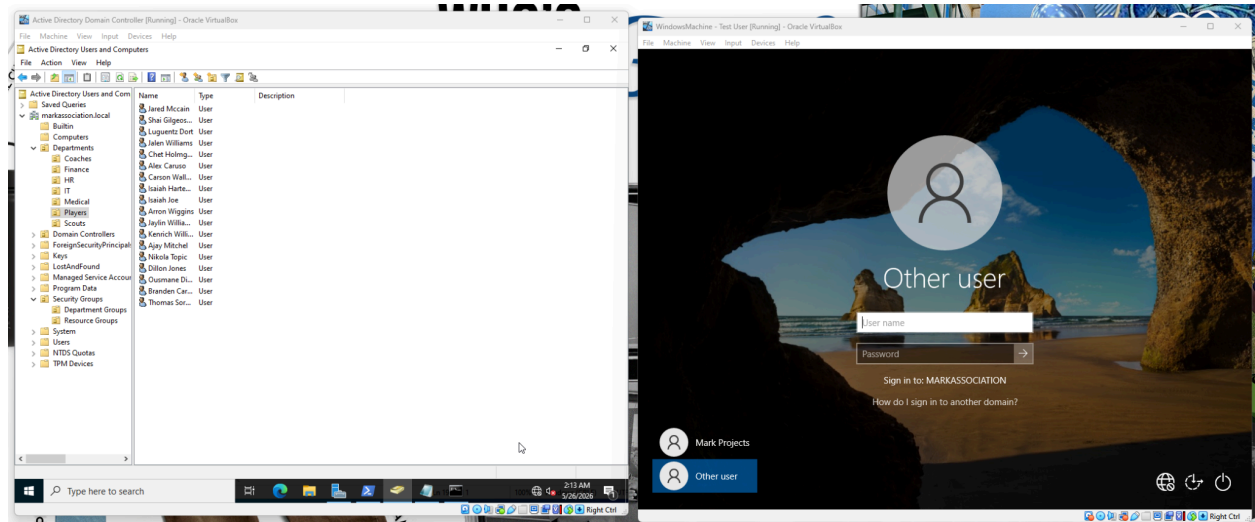


After providing the correct credentials here is what it says

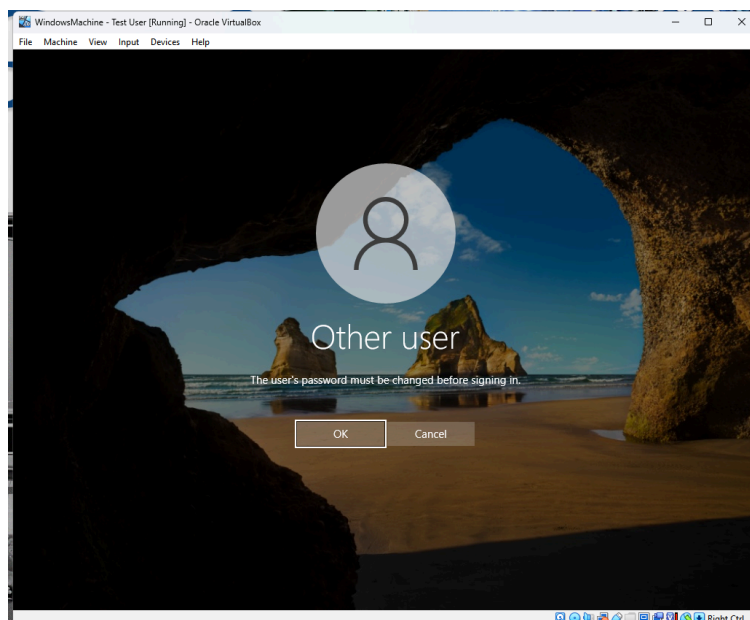


I then have to restart my VM

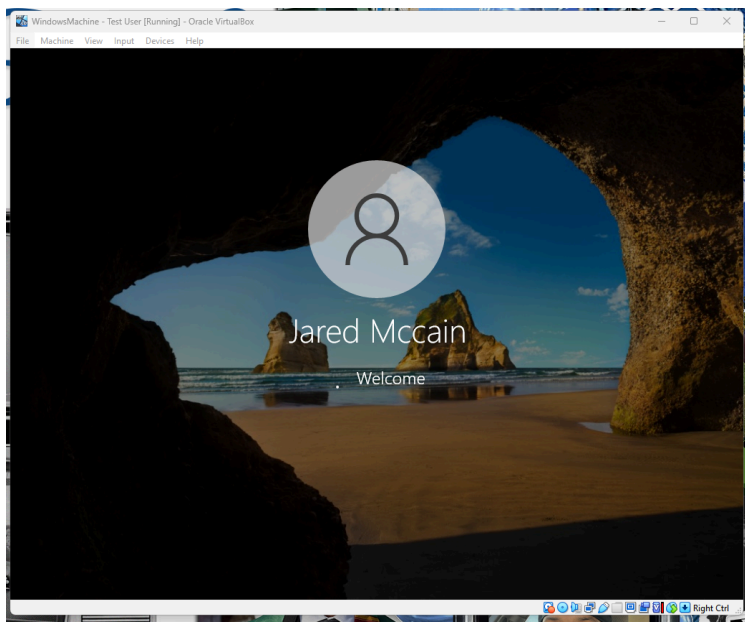
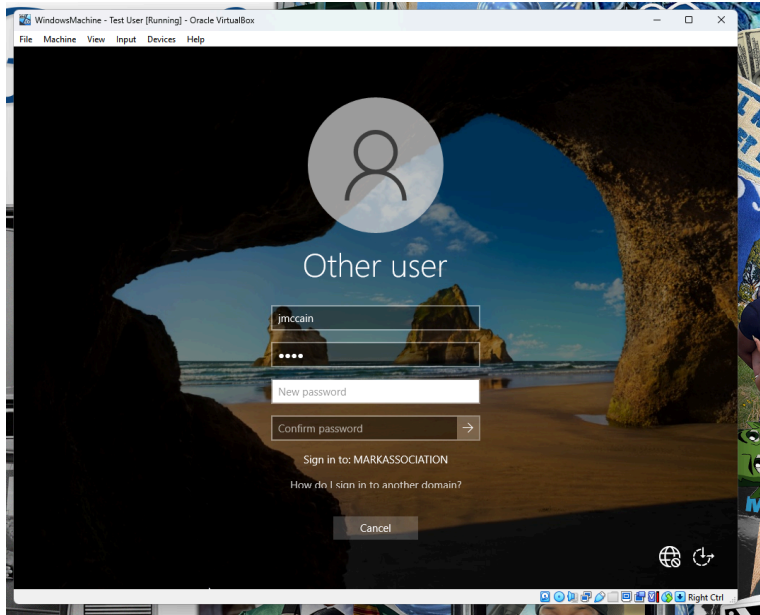
Now I can finally login with one of the users's I have created and when I look under Computers the Workstation will be added to the Domain Controller



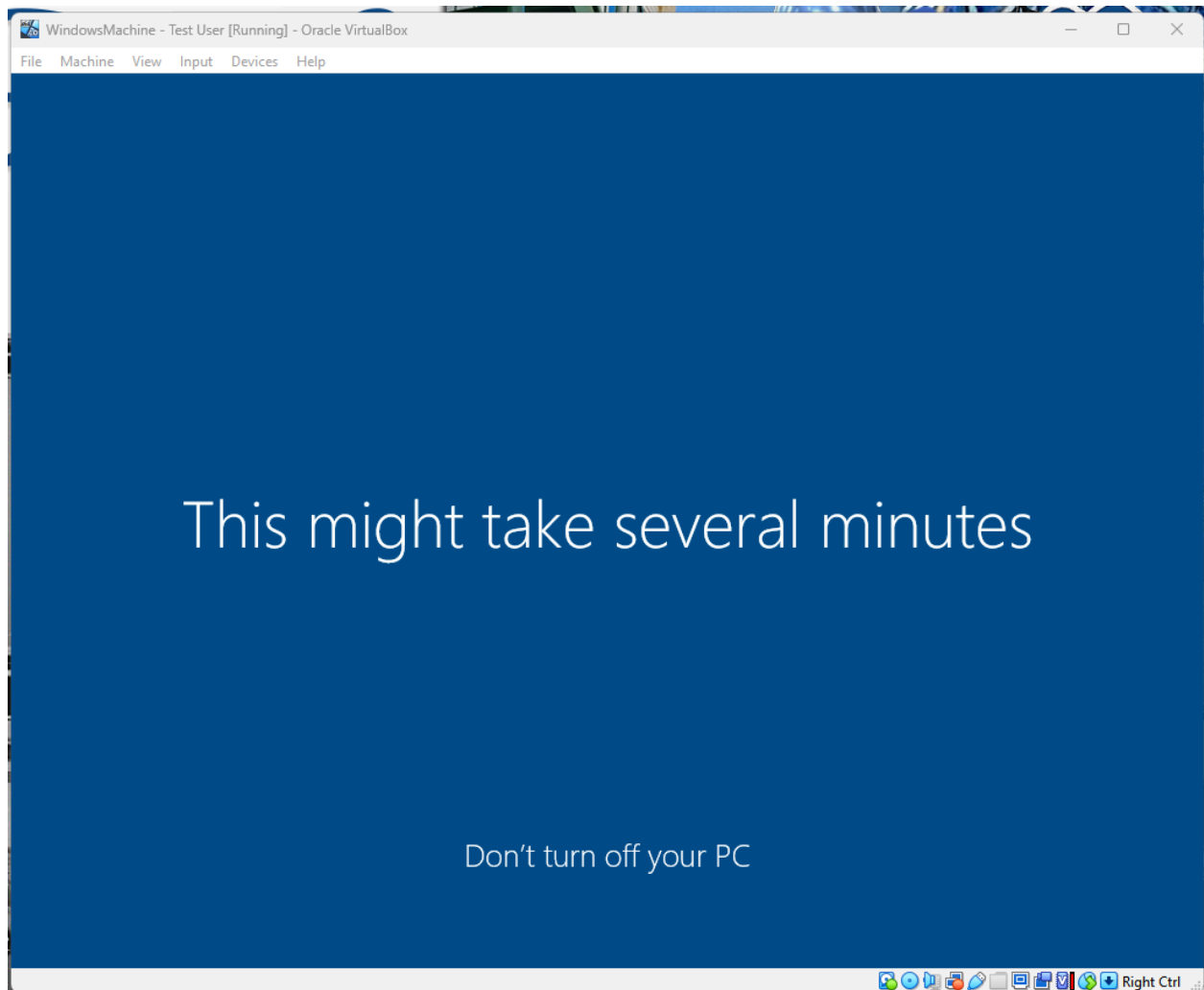
I will sign in with Jarred Mccain  
User: JMccain



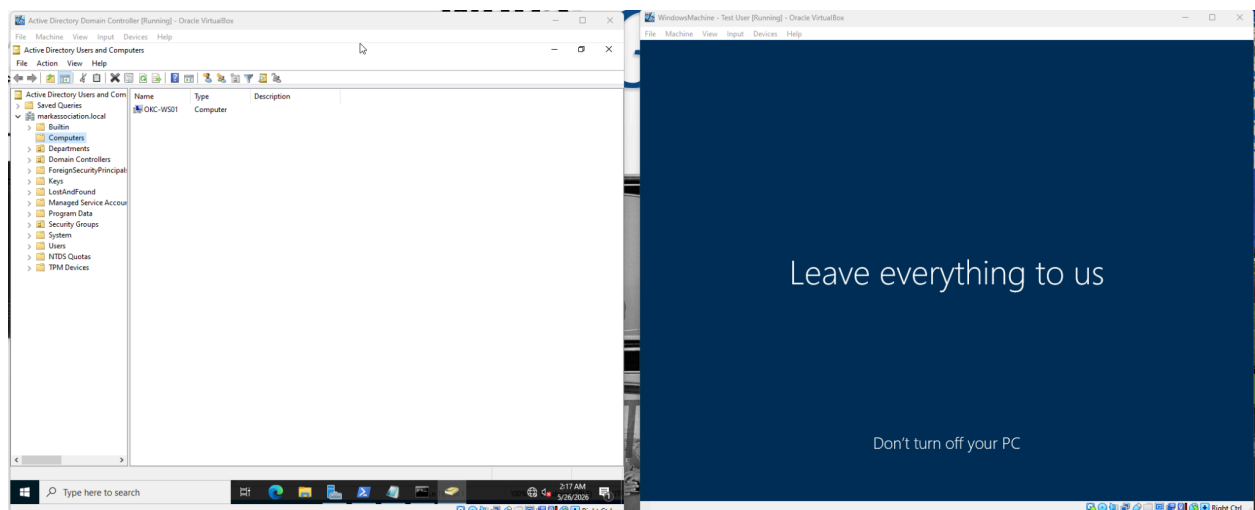
- In my Domain Controller - Group Policy Object, I do have the setting checked stating user must change their password upon first login



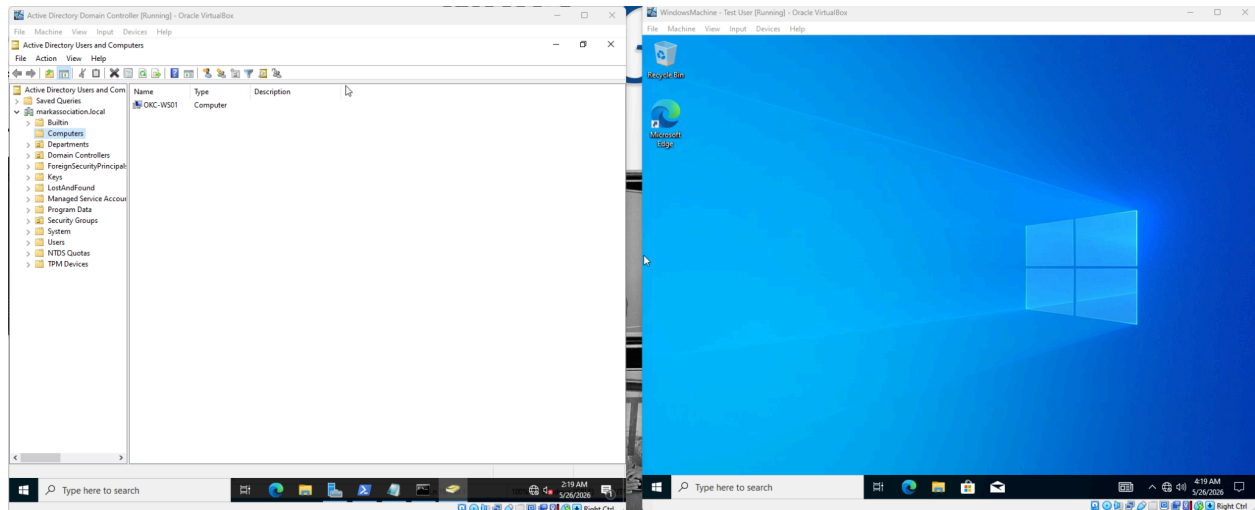
I'm now changing the password and the laptop (VM) is logging in



It will go through the usual pc set up process



Now to the left you can see that the computer is showing in my domain controller



We now have successfully created:

- Installed & Configured an Active Directory Domain Controller
- Installed & Configured an Entra ID & Intune
- Created Users, Organizational Units (OUs), Security Groups, Computers, Group Policy Object (GPOs), RBAC,
- Network Protocol Configuration - Configuring an Internal Network
- Domain-Joined Workstation,
- IP address, DNS server